IMO 2026 performance by ChatGPT 5.6 Pro

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Abstract

We present one-shot solution attempts by ChatGPT 5.6 Pro on the six IMO 2026 problems. The attempts were made through the publicly available ChatGPT web app on the day of the contest, with web search disabled. The 41/42 score is the authors' assessment, based on AI-assisted and manual review. The reevaluation of Problem 2 was assisted by ChatGPT 5.6 Pro. Reported solution times are the elapsed times displayed by the ChatGPT app.

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Problem numbers, names, and page numbers link to the corresponding problem. The names are descriptive navigation labels, not official IMO titles. Elapsed times are taken from the ChatGPT app's UI display. The authors' assessment column records the authors' own scores; these are not official IMO jury scores. Gemini scores link to the full Gemini reviews in Appendix A; the revised Problem 2 authors' score links to the additional jury-style review there.

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Problem 1: GCD–LCM Blackboard Process

Area	Number theory	**Elapsed time**	2m 38s
Transcript	Open shared ChatGPT transcript		
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Problem. *There are* 2026 *integers greater than* 1 *written on a blackboard, not necessarily distinct. In one move, Confucius chooses two integers $m > 1$ and $n > 1$ written in different positions and replaces them by*

$$\gcd(m,n) \qquad \textit{and} \qquad \frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}.$$

He continues making moves for as long as a move is possible.

(a) Prove that, regardless of his choices, the process terminates after finitely many moves with exactly one integer $M > 1$ on the board.

(b) Prove that the value of M is independent of all choices made during the process.

Solution

Proof. Write the entries currently on the board as

$$x_1, x_2, \dots, x_{2026}.$$

Entries equal to 1 remain on the board, but they cannot be selected for a subsequent move.

Part (a): termination and the final number of nontrivial entries.

For every positive integer x, let

$$\Omega(x) = \sum_p v_p(x),$$

where the sum is over all primes and $v_p(x)$ is the exponent of p in the prime factorization of x. Thus $\Omega(1) = 0$. At any stage of the process, define

$$T = \sum_{i=1}^{2026} \Omega(x_i) \qquad \text{and} \qquad R = \#\{i : x_i > 1\}.$$

We shall prove that the nonnegative integer

$$\Phi = T + R$$

strictly decreases after every move.

Suppose a move is made using $m, n > 1$, and put

$$d = \gcd(m,n), \qquad q = \frac{\operatorname{lcm}(m,n)}{d}.$$

Fix a prime p, and write

$$a = v_p(m), \qquad b = v_p(n).$$

Then

$$v_p(d) = \min(a,b)$$

and

$$v_p(q) = v_p(\mathrm{lcm}(m,n)) - v_p(d) = \max(a,b) - \min(a,b) = |a-b|.$$

Consequently,

$$\begin{aligned}(v_p(m) + v_p(n)) - (v_p(d) + v_p(q)) &= a + b - \min(a,b) - |a-b| \\ &= \min(a,b).\end{aligned}$$

Summing over all primes gives

$$T_{\text{old}} - T_{\text{new}} = \sum_p \min(v_p(m), v_p(n)) = \Omega(d).$$

There are now two cases.

If $d > 1$, then $\Omega(d) \geq 1$, so T strictly decreases. The two selected entries are replaced by at most two entries greater than 1, so R does not increase. Hence Φ strictly decreases.

If $d = 1$, then T is unchanged. Since m and n are coprime,

$$q = \mathrm{lcm}(m,n) = mn > 1.$$

Thus the two selected entries greater than 1 are replaced by exactly one entry greater than 1 and one entry equal to 1. Therefore R decreases by exactly 1, and again Φ strictly decreases.

Since Φ is a nonnegative integer, it cannot decrease indefinitely. The process therefore terminates after finitely many moves.

At termination there cannot be two entries greater than 1, since any such pair could be selected for another move. Hence $R \leq 1$. On the other hand, a move can never remove all entries greater than 1: if $d > 1$, then the new entry d is greater than 1, while if $d = 1$, then the new entry $q = mn$ is greater than 1. Since initially every entry is greater than 1, we always have $R \geq 1$. Thus the terminal board contains exactly one integer $M > 1$.

Part (b): determination and uniqueness of the final value.

Fix a prime p. For a board with entries $x_1, \ldots, x_{2026}$, define

$$I_p = \gcd(v_p(x_1), v_p(x_2), \ldots, v_p(x_{2026})),$$

with the conventions $\gcd(0,t) = t$ and $\gcd(0,\ldots,0) = 0$. We claim that I_p is invariant under every move.

Indeed, suppose the two selected entries have p-adic valuations a and b. As shown above, after the move these two valuations become

$$\min(a,b) \qquad \text{and} \qquad |a-b|.$$

Moreover,

$$\gcd(\min(a,b), |a-b|) = \gcd(a,b).$$

For example, if $a \leq b$, then

$$\gcd(\min(a,b), |a-b|) = \gcd(a, b-a) = \gcd(a,b),$$

and the case $b \leq a$ is symmetric. Since all other p-adic valuations are unchanged, the gcd of the entire list of valuations is unchanged. Therefore I_p is an invariant.

Let the initial entries be $a_1, \ldots, a_{2026}$. On the terminal board, all entries except M are equal to 1. Hence its list of p-adic valuations is

$$v_p(M), 0, \ldots, 0,$$

so the terminal value of the invariant is

$$I_p = v_p(M).$$

Comparing with its initial value yields, for every prime p,

$$v_p(M) = \gcd\big(v_p(a_1), v_p(a_2), \ldots, v_p(a_{2026})\big).$$

It follows that

$$\boxed{M = \prod_{p \mid a_1 a_2 \cdots a_{2026}} p^{\gcd(v_p(a_1), v_p(a_2), \ldots, v_p(a_{2026}))}}.$$

Every exponent in this prime factorization is determined solely by the initial board. By uniqueness of prime factorization, the final value M is therefore independent of all choices made by Confucius. $\square$

Problem 2: Equal Midpoint Distances from a Circumcentre

Area	Geometry	**Elapsed time**	22m 37s
Transcript	Open shared ChatGPT transcript		
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Problem. *Let ABC be a triangle, and let M and N be the midpoints of AB and AC, respectively. Let K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside $\angle LBA$, L lies inside $\angle ACK$, and*

$$\angle KBA = \angle ACL, \qquad \angle LBK = \angle LNC, \qquad \angle LCK = \angle BMK.$$

If O is the circumcentre of triangle AKL, then

$$OM = ON.$$

Solution

Proof. Let ω be the circumcircle of AKL. We shall prove that M and N have equal powers with respect to ω. Since

$$\operatorname{Pow}_\omega(X) = OX^2 - R_\omega^2$$

for every point X, equality of those powers will immediately imply $OM = ON$.

Set

$$x = \angle KBA = \angle ACL, \qquad y = \angle LBK = \angle LNC, \qquad z = \angle LCK = \angle BMK.$$

Because BK lies inside $\angle LBA$ and CL lies inside $\angle ACK$,

$$\angle LBA = x + y, \qquad \angle ACK = x + z. \tag{1}$$

1. The order of the rays through A. We first show that the rays from A occur in the order

$$AB,\ AK,\ AL,\ AC. \tag{2}$$

Indeed, write the positive barycentric coordinates of K and L with respect to ABC as

$$K = (a_K : b_K : c_K), \qquad L = (a_L : b_L : c_L).$$

For an interior point $P = (a : b : c)$, the position of the ray BP inside $\angle ABC$, measured from BA toward BC, is determined monotonically by c/a. Since BK lies between BA and BL,

$$\frac{c_K}{a_K} < \frac{c_L}{a_L}. \tag{3}$$

Similarly, the position of CP inside $\angle ACB$, measured from CA toward CB, is determined monotonically by b/a. Since CL lies between CA and CK,

$$\frac{b_L}{a_L} < \frac{b_K}{a_K}. \tag{4}$$

Consequently,

$$\frac{c_K}{b_K} = \frac{c_K/a_K}{b_K/a_K} < \frac{c_L/a_L}{b_L/a_L} = \frac{c_L}{b_L}.$$

Since the position of AP inside $\angle BAC$, measured from AB toward AC, is determined monotonically by c/b, this proves (2).

Define

$$p = \angle BAK, \qquad r = \angle CAL, \qquad \theta = \angle KAL.$$

Thus

$$\angle BAC = p + \theta + r. \tag{5}$$

Also set

$$\kappa = \angle AKL, \qquad \lambda = \angle ALK, \qquad S = \kappa + \lambda = \pi - \theta.$$

Finally, write

$$c = AB, \qquad d = AC, \qquad \rho = \frac{d}{c}.$$

2. Relations forced by the two midpoint conditions. In triangle BMK, we have

$$BM = \frac{c}{2}, \qquad \angle MBK = x, \qquad \angle BMK = z.$$

The sine rule therefore gives

$$BK = \frac{c}{2}\frac{\sin z}{\sin(x+z)}.$$

On the other hand, the sine rule in triangle ABK gives

$$BK = c\frac{\sin p}{\sin(x+p)}.$$

Hence

$$\sin(x+p)\sin z = 2\sin p \sin(x+z). \tag{6}$$

The same argument in triangles CNL and ACL yields

$$\sin(x+r)\sin y = 2\sin r \sin(x+y). \tag{7}$$

For brevity, put

$$X = \sin x, \quad Y = \sin(x+y), \quad Z = \sin(x+z), \quad P = \sin(x+p), \quad Q = \sin(x+r). \tag{8}$$

The remaining angles in triangles ABL and ACK are

$$\phi := \angle ALB = S - p - x - y, \qquad \psi := \angle AKC = S - r - x - z. \tag{9}$$

The sine rule in triangles ACL and ABL gives

$$AL = \frac{dX}{Q} = \frac{cY}{\sin\phi},$$

whereas the sine rule in triangles ABK and ACK gives

$$AK = \frac{cX}{P} = \frac{dZ}{\sin\psi}.$$

Consequently,

$$\rho = \frac{YQ}{X\sin\phi} = \frac{X\sin\psi}{PZ}. \tag{10}$$

Equating the two expressions in (10), we obtain

$$X^2 \sin\phi \sin\psi = YPQZ. \tag{11}$$

Moreover, the sine rule in triangle AKL gives

$$\frac{AK}{AL} = \frac{\sin\lambda}{\sin\kappa}.$$

Using the expressions above for AK and AL, we find

$$\frac{\sin\kappa}{\sin\lambda} = \frac{\rho P}{Q}. \tag{12}$$

3. The powers of M and N. We use directed lengths on secants and directed angles modulo π. Let $T \neq K$ be the second intersection of the line MK with ω, and orient that line from M toward K. Since A, K, L, T are concyclic,

$$\measuredangle ATM \equiv \measuredangle ATK \equiv \measuredangle ALK = \lambda \pmod{\pi}.$$

Also, because MA and MB are opposite rays and $\angle BMK = z$,

$$\measuredangle AMT \equiv \pi - z \pmod{\pi}.$$

Thus the directed sine rule in triangle AMT gives

$$\frac{\overline{MT}}{MA} = \frac{\sin(z-\lambda)}{\sin\lambda}.$$

Furthermore, the sine rule in triangle BMK gives

$$MK = \frac{cX}{2Z}.$$

Therefore, by the secant form of the power theorem,

$$\mathrm{Pow}_\omega(M) = \overline{MK}\,\overline{MT} = \frac{c^2X}{4Z}\frac{\sin(z-\lambda)}{\sin\lambda}. \tag{13}$$

Likewise, if $U \neq L$ is the second intersection of the line NL with ω, then

$$\mathrm{Pow}_\omega(N) = \frac{d^2X}{4Y}\frac{\sin(y-\kappa)}{\sin\kappa}. \tag{14}$$

Thus $\mathrm{Pow}_\omega(M) = \mathrm{Pow}_\omega(N)$ is equivalent to

$$\frac{\sin(\lambda - z)}{Z\sin\lambda} = \rho^2\frac{\sin(\kappa - y)}{Y\sin\kappa}. \tag{15}$$

4. Trigonometric reduction. Put

$$t = \frac{\sin\kappa}{\sin\lambda} = \frac{\rho P}{Q}, \tag{16}$$

where the second equality is (12). Since $S = \kappa + \lambda$, the identities

$$\sin S\,\sin(\lambda - z) = \sin\lambda\,\sin(S - z) - \sin\kappa\,\sin z$$

and

$$\sin S\,\sin(\kappa - y) = \sin\kappa\,\sin(S-y) - \sin\lambda\,\sin y$$

give

$$\frac{\sin(\lambda - z)}{\sin\lambda} = \frac{\sin(S-z) - t\sin z}{\sin S}, \tag{17}$$

$$\frac{\sin(\kappa - y)}{\sin\kappa} = \frac{\sin(S-y) - t^{-1}\sin y}{\sin S}. \tag{18}$$

Substituting (16)–(18) into (15), and then using (6)–(7), shows that (15) is equivalent to $D = 0$, where

$$D = \frac{\sin(S-z)}{Z} - \rho^2\frac{\sin(S-y)}{Y} - 2\rho\left(\frac{\sin p}{Q} - \frac{\sin r}{P}\right). \tag{19}$$

Multiplying by $PZ\sin\phi$ and using (10), we obtain

$$PZ\sin\phi\, D = E(S), \tag{20}$$

where

$$E(S) = P\sin\phi\sin(S-z) - Q\sin\psi\sin(S-y) + \frac{2YZ}{X}(Q\sin r - P\sin p). \tag{21}$$

We claim that

$$E(S) = \frac{\sin(p-r)}{X^2}\left(X^2\sin\phi\sin\psi - YPQZ\right). \tag{22}$$

To prove the claim, define

$$F(S) = E(S) - \frac{\sin(p-r)}{X^2}\left(X^2\sin\phi\sin\psi - YPQZ\right)$$

and put

$$T = 2S - p - r - 2x - y - z.$$

By the product-to-sum formulas, the entire S-dependent part of $2F(S)$ is

$$-P\cos(T + x + r) + Q\cos(T + x + p) + \sin(p-r)\cos T. \tag{23}$$

Now $P = \sin(x+p)$ and $Q = \sin(x+r)$, so

$$\begin{aligned} &Q\cos(T+x+p) - P\cos(T+x+r)\\ &\quad= \cos T\big(Q\cos(x+p) - P\cos(x+r)\big)\\ &\qquad - \sin T\big(Q\sin(x+p) - P\sin(x+r)\big)\\ &\quad= -\sin(p-r)\cos T. \end{aligned}$$

This cancels the last term in (23). Hence $F(S)$ is independent of S.

We may therefore evaluate it at the convenient value

$$S_0 = x + y + z.$$

Dividing (6) by $\sin p\sin z$, and (7) by $\sin r\sin y$, respectively, gives

$$\cot p = \cot x + 2\cot z, \qquad \cot r = \cot x + 2\cot y.$$

It follows that

$$\sin(z-p) = \frac{\sin p\, Z}{X}, \qquad \sin(y-r) = \frac{\sin r\, Y}{X}. \tag{24}$$

At $S = S_0$, we have $\phi = z - p$ and $\psi = y - r$. Using (24) in (21), we get

$$E(S_0) = \frac{YZ}{X}(Q\sin r - P\sin p), \tag{25}$$

while

$$X^2 \sin\phi \sin\psi - YPQZ = YZ(\sin p \sin r - PQ). \tag{26}$$

Finally, the elementary identities

$$\begin{aligned} Q\sin r - P\sin p &= -\sin(x+p+r)\sin(p-r), \\ \sin p \sin r - PQ &= -X\sin(x+p+r) \end{aligned}$$

show that

$$X(Q\sin r - P\sin p) = \sin(p-r)(\sin p\sin r - PQ).$$

Together with (25) and (26), this gives $F(S_0) = 0$. Since F is constant, $F(S) = 0$ for all S, and therefore (22) is proved.

Now (11) makes the right-hand side of (22) equal to zero. Hence $E(S) = 0$. By (20), and because $PZ\sin\phi \neq 0$, we obtain $D = 0$. Therefore (15) holds, and (13)–(14) give

$$\mathrm{Pow}_\omega(M) = \mathrm{Pow}_\omega(N).$$

Consequently,

$$OM^2 - R_\omega^2 = ON^2 - R_\omega^2,$$

so $OM = ON$, as required. ☐

Problem 3: Mark-and-Claim Stick Game

Area	Combinatorics	**Elapsed time**	13m 19s
Transcript	Open shared ChatGPT transcript		
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Problem. *Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1. First Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n further points; all marked points are distinct. The stick is cut at every marked point. The players then alternately claim unclaimed pieces, with Liu Bang moving first, and each player seeks to maximize the total length of the pieces he claims. Determine the largest number c that Liu Bang can guarantee, regardless of Xiang Yu's play.*

Solution

Theorem 1. *For every positive integer n, the required value is*

$$\boxed{c_n = \frac{2^n}{2^{n+1}-1}}.$$

We begin with three lemmas.

Lemma 1 (Value of the claiming phase)**.** *Suppose the final piece lengths, in nonincreasing order, are*

$$x_1 \geq x_2 \geq \cdots \geq x_m > 0.$$

Then the value of the claiming phase for the first player is

$$x_1 + x_3 + x_5 + \cdots .$$

Equivalently, if

$$\mathrm{Alt}(x_1, \ldots, x_m) := x_1 - x_2 + x_3 - x_4 + \cdots ,$$

then the first player receives, under optimal play,

$$\frac{1 + \mathrm{Alt}(x_1, \ldots, x_m)}{2}.$$

Proof. If the first player always takes a largest remaining piece, then just before his jth move at most $2j-2$ pieces have been removed. Hence at least one of the original $2j-1$ largest pieces remains, so the piece he takes on that move has length at least x_{2j-1}. Thus he can guarantee

$$x_1 + x_3 + x_5 + \cdots .$$

Conversely, suppose the second player always takes a largest remaining piece. Just before his jth move, exactly $2j-1$ pieces have been removed, so at least one of the original $2j$ largest pieces remains. He therefore takes a piece of length at least x_{2j}. Hence the second player can guarantee at least

$$x_2 + x_4 + x_6 + \cdots ,$$

leaving the first player at most the complementary sum $x_1 + x_3 + x_5 + \cdots$. This proves the first assertion. Since the total length is 1, the second formula follows. ☐

Lemma 2 (Cancellation of equal pairs)**.** *Suppose a multiset of positive numbers consists of some equal pairs, together with a residual multiset of total sum d. If all the numbers are arranged in nonincreasing order, then their alternating sum is at most d.*

Proof. In a nonincreasing list, equal entries occur consecutively. Deleting two adjacent equal entries does not change the alternating sum: their contributions cancel, and every later index shifts by 2, preserving its parity. Deleting all prescribed equal pairs therefore leaves the alternating sum of the residual entries alone. For a nonincreasing residual list $z_1 \geq \cdots \geq z_s > 0$, this alternating sum is

$$(z_1 - z_2) + (z_3 - z_4) + \cdots ,$$

with an additional final term z_s when s is odd. It is nonnegative and no larger than $z_1 + \cdots + z_s = d$. ☐

Lemma 3 (Matching two families of intervals)**.** *Let one family consist of p intervals of total length A, and a second nonempty family consist of q intervals of total length B, where $A \geq B$. Using at most $p+q-1$ cuts in these intervals, one can subdivide them so that every fragment from the second family is matched by an equal-length fragment from the first family, while the unmatched fragments from the first family have total length $A - B$.*

Proof. Order the intervals arbitrarily within each family and concatenate them conceptually, obtaining intervals $[0, A]$ and $[0, B]$. On the common portion $[0, B]$, refine both concatenation partitions by taking the union of their boundary points. The resulting subinterval lengths on $[0, B]$ are identical for the two families.

To realize this common refinement in the original intervals, on the first side it is enough to insert the $q - 1$ internal boundaries of the second concatenation and, if necessary, the point B. This uses at most q cuts. On the second side, it is enough to insert those internal boundaries of the first concatenation that lie in $(0, B)$, using at most $p - 1$ cuts. Thus at most $p + q - 1$ cuts are needed. The part of the first concatenation lying beyond B has total length $A - B$ and is precisely the unmatched part. ☐

Proof of the theorem. Set

$$u := \frac{1}{2^{n+1} - 1}.$$

We prove matching upper and lower bounds.

Upper bound. We show that, after any choice by Liu Bang, Xiang Yu can arrange that

$$\mathrm{Alt}(x_1, \ldots, x_m) \leq u. \tag{1}$$

By the first lemma, this will hold Liu Bang to at most

$$\frac{1+u}{2} = \frac{2^n}{2^{n+1} - 1}. \tag{2}$$

Suppose first that Liu Bang uses $r < n$ marks, producing $r + 1 \leq n$ pieces. Xiang Yu bisects each of these pieces. Every final length then occurs in an equal pair, so the second lemma gives $\mathrm{Alt}(x_1, \ldots, x_m) = 0$. Thus it remains only to consider the case in which Liu Bang uses exactly n marks and creates $n + 1$ initial pieces. Let their lengths be

$$a_1, a_2, \ldots, a_{n+1}.$$

Consider the 2^{n+1} subset sums

$$s(S) := \sum_{i \in S} a_i, \qquad S \subseteq \{1, 2, \ldots, n+1\}.$$

They all lie in $[0, 1]$, and the smallest and largest are 0 and 1. When these subset sums are listed in nondecreasing order, the $2^{n+1} - 1$ consecutive gaps have total sum 1. Hence two distinct subsets S, T satisfy

$$|s(S) - s(T)| \leq \frac{1}{2^{n+1} - 1} = u. \tag{3}$$

Delete the common indices and put

$$P := S \setminus T, \qquad Q := T \setminus S.$$

After interchanging P and Q if necessary, we may assume

$$\sum_{i \in P} a_i \geq \sum_{i \in Q} a_i.$$

Define

$$d := \sum_{i \in P} a_i - \sum_{i \in Q} a_i.$$

Then P and Q are disjoint, not both empty, and by (3),

$$0 \leq d \leq u. \tag{4}$$

Let

$$R := \{1, 2, \ldots, n+1\} \setminus (P \cup Q).$$

If Q is nonempty, apply the matching lemma to the pieces indexed by P and Q. It uses at most $|P| + |Q| - 1$ cuts and produces equal pairs of fragments, except for unmatched fragments on the P-side whose total length is d. Xiang Yu also bisects every piece indexed by R, using $|R|$ additional cuts. The total number of cuts is at most

$$|P| + |Q| - 1 + |R| = n.$$

If Q is empty, Xiang Yu simply leaves the pieces indexed by P unmatched and bisects every piece indexed by R. Since P is nonempty, this uses

$$|R| = n + 1 - |P| \leq n$$

cuts, and the total length of the unmatched pieces is again d.

In either case, all final pieces except for a residual collection of total length d occur in equal pairs. The cancellation lemma and (4) therefore give

$$\mathrm{Alt}(x_1, \ldots, x_m) \leq d \leq u,$$

which proves (1) and hence the desired upper bound.

Lower bound. Liu Bang marks the points

$$(2^k - 1)u, \qquad k = 1, 2, \ldots, n. \tag{5}$$

The resulting $n+1$ initial pieces have lengths

$$u, 2u, 4u, \dots, 2^n u, \tag{6}$$

whose sum is

$$(1 + 2 + \cdots + 2^n)u = (2^{n+1} - 1)u = 1.$$

We show that, whatever cuts Xiang Yu makes, the final alternating sum satisfies

$$\operatorname{Alt}(x_1, \dots, x_m) \ge u. \tag{7}$$

Let Xiang Yu introduce $k \le n$ cuts. There are then

$$m = n + 1 + k \le 2n + 1$$

final pieces. Arrange their lengths in nonincreasing order,

$$x_1 \ge x_2 \ge \cdots \ge x_m > 0,$$

breaking ties arbitrarily, and pair them as

$$(x_1, x_2),\ (x_3, x_4),\ \dots.$$

If m is odd, leave x_m unpaired. Thus

$$\operatorname{Alt}(x_1, \dots, x_m) = \sum_j (x_{2j-1} - x_{2j}) + \begin{cases} x_m, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \tag{8}$$

Construct a multigraph G with one vertex for each of the $n+1$ initial pieces in (6). Every final fragment lies in a unique initial piece. For each paired pair of final fragments, draw an edge between the two vertices corresponding to their initial pieces; a pair whose fragments come from the same initial piece produces a loop. Parallel edges are allowed. If m is odd, also remember the vertex containing the unpaired fragment x_m.

The graph has $n+1$ vertices and

$$\left\lfloor \frac{m}{2} \right\rfloor \le n$$

edges. Hence at least one connected component of G is a tree. Indeed, a connected multigraph that is not a tree has at least as many edges as vertices, so if no component were a tree, the whole graph would have at least $n+1$ edges.

Choose a tree component C. Since C is bipartite, assign signs $\varepsilon_v \in \{+1, -1\}$ to its vertices so that adjacent vertices have opposite signs. Let a_v denote the length of the initial piece corresponding to v.

Every paired fragment arising from a vertex of C is paired with a fragment from another vertex of C, because C is a connected component. For an edge $e = vw$ of C, let $\lambda_{e,v}$ and $\lambda_{e,w}$ be the lengths of its two endpoint fragments. Since $\varepsilon_w = -\varepsilon_v$, the contribution of these two fragments to the signed sum is

$$\varepsilon_v \lambda_{e,v} + \varepsilon_w \lambda_{e,w} = \varepsilon_v (\lambda_{e,v} - \lambda_{e,w}).$$

Its absolute value is exactly the difference between the two lengths in the corresponding ranked pair. If the unpaired fragment x_m comes from a vertex of C, it contributes $\pm x_m$. Summing over the component and applying the triangle inequality to (8), we obtain

$$\left| \sum_{v \in C} \varepsilon_v a_v \right| \le \operatorname{Alt}(x_1, \dots, x_m). \tag{9}$$

The numbers a_v are distinct members of

$$u, 2u, 4u, \ldots, 2^n u.$$

Let $2^J u$ be the largest one represented in C. Its absolute contribution to the signed sum in (9) is $2^J u$, whereas the sum of all possible smaller contributions is at most

$$(1 + 2 + \cdots + 2^{J-1})u = (2^J - 1)u.$$

Consequently,

$$\left| \sum_{v \in C} \varepsilon_v a_v \right| \geq 2^J u - (2^J - 1)u = u. \tag{10}$$

Combining (9) and (10) proves (7).

By the claiming-phase lemma, Liu Bang can therefore guarantee at least

$$\frac{1+u}{2} = \frac{2^n}{2^{n+1} - 1}.$$

Together with the upper bound, this completes the proof. □

Problem 4: Triangle-Cutting Angle Game

Area	Geometry	**Elapsed time**	5m 22s
Transcript	Open shared ChatGPT transcript		
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Problem. *Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?*

Solution

Let

$$\Omega = 180^\circ.$$

We determine exactly for which values of $\theta \in (0^\circ, 180^\circ)$ Mulan can force the game to end after finitely many cuts.

Theorem 1. *Mulan can guarantee victory if and only if*

$$\boxed{\theta = \frac{180^\circ}{n}}$$

for some integer $n \geq 2$.

Proof. We prove the two directions separately.

A preliminary claim.

Claim 1. *Suppose the current triangle contains an angle equal to $m\theta$, where m is a positive integer. Then Mulan can force a win in at most $m-1$ further cuts.*

Proof. We use induction on m. If $m = 1$, the triangle already has an angle equal to θ, so the game has ended and Mulan has won.

Now let $m \geq 2$, and suppose one angle of the current triangle is $m\theta$. From the vertex of that angle, Mulan draws an interior ray that makes an angle θ with one of the two adjacent sides. The ray meets the opposite side at an interior point and splits the angle $m\theta$ into

$$\theta \qquad \text{and} \qquad (m-1)\theta.$$

Thus one of the two resulting triangles has an angle θ, while the other has an angle $(m-1)\theta$. If Shan-Yu keeps the first triangle, Mulan wins immediately. If he keeps the second, the induction hypothesis applies. Hence Mulan wins after at most $m-1$ cuts. □

Sufficiency.

Assume that

$$\theta = \frac{\Omega}{n}$$

for some integer $n \geq 2$. Consider any current triangle, and write its three angles as

$$a\theta, \qquad b\theta, \qquad c\theta,$$

where

$$a, b, c > 0 \qquad \text{and} \qquad a + b + c = n.$$

If one of a, b, c is an integer, then the preliminary claim already shows that Mulan can force a win. It remains to consider the case in which none of a, b, c is an integer.

We use the following elementary lemma.

Lemma 1. *Let $a, b, c > 0$ be nonintegers whose sum is an integer $n \geq 2$. After relabeling a, b, c, there is an integer k such that*

$$b < k < a + b.$$

Proof. First suppose that one of the three numbers is greater than 1. Call that number a, choose either of the remaining numbers as b, and set

$$k = \lceil b \rceil.$$

Because b is not an integer,

$$0 < k - b < 1 < a,$$

so indeed $b < k < a + b$.

Now suppose that all three numbers are less than 1. Their sum is an integer at least 2 and strictly less than 3, so their sum is exactly 2. Let a and b be the two largest of the three numbers, and let the remaining number be c. Then

$$b < 1 \qquad \text{and} \qquad a + b = 2 - c > 1.$$

Thus $b < 1 < a + b$, and we may take $k = 1$. $\square$

Label the vertices of the current triangle as A, B, C so that

$$\angle A = a\theta, \qquad \angle B = b\theta, \qquad \angle C = c\theta,$$

and choose an integer k satisfying

$$b < k < a + b.$$

From A, Mulan draws the interior ray meeting BC at P and satisfying

$$\angle BAP = (k - b)\theta.$$

This is a legal cut because

$$0 < k - b < a.$$

In triangle ABP, the angles at A and B have sum

$$(k - b)\theta + b\theta = k\theta.$$

Therefore

$$\angle APB = \Omega - k\theta = (n-k)\theta.$$

Since B, P, C are collinear, the two angles at P are supplementary, and hence

$$\angle APC = \Omega - (n-k)\theta = k\theta.$$

Thus triangle ABP contains the positive integral multiple $(n-k)\theta$, while triangle ACP contains the positive integral multiple $k\theta$. Notice that

$$0 < b < k < a + b = n - c < n,$$

so $1 \leq k \leq n-1$, and both multiples are positive.

Whichever triangle Shan-Yu keeps, the preliminary claim applies. Therefore Mulan can force a win in finitely many cuts. In fact, the argument gives a uniform bound of at most $n-1$ cuts.

Necessity.

Now suppose that

$$\frac{\Omega}{\theta} \notin \mathbb{Z}.$$

Define the additive subgroup

$$H = \{m\theta : m \in \mathbb{Z}\}$$

of the real numbers. The assumption says precisely that

$$\Omega \notin H.$$

Shan-Yu begins with an equilateral triangle. Each of its angles is $\Omega/3$, and $\Omega/3 \notin H$: otherwise $\Omega/3 = m\theta$ for some integer m, which would imply $\Omega = 3m\theta \in H$, a contradiction.

Shan-Yu will maintain the following invariant:

none of the three angles of the current triangle belongs to H.

Suppose the current triangle is ABC, with

$$\angle A = a, \qquad \angle B = b, \qquad \angle C = c,$$

and suppose $a, b, c \notin H$. Mulan chooses a point P in the interior of BC and cuts along AP. Put

$$x = \angle BAP,$$

so that $0 < x < a$. The two resulting triangles have angle triples

$$ABP: \qquad x,\ b,\ \Omega - b - x,$$

and

$$ACP: \qquad a - x,\ c,\ b + x.$$

We claim that at least one of these two triangles has no angle in H. Suppose, to the contrary, that each resulting triangle has an angle in H. Since $b, c \notin H$, one of

$$x, \qquad \Omega - b - x$$

must belong to H, and one of

$$a - x, \qquad b + x$$

must belong to H. There are four possible pairings:

$$\begin{aligned} x,\ a-x\in H &\implies a = x+(a-x)\in H,\\ x,\ b+x\in H &\implies b=(b+x)-x\in H,\\ \Omega-b-x,\ a-x\in H &\implies c=(\Omega-b-x)-(a-x)\in H,\\ \Omega-b-x,\ b+x\in H &\implies \Omega=(\Omega-b-x)+(b+x)\in H. \end{aligned}$$

Each conclusion is impossible. Therefore at least one of the two resulting triangles has all three of its angles outside H. Shan-Yu keeps such a triangle, preserving the invariant.

By repeating this strategy after every cut, Shan-Yu can ensure forever that no angle of the current triangle lies in H. Since $\theta \in H$, the current triangle never has an angle equal to θ. Thus Mulan cannot guarantee victory.

Combining the two directions, Mulan has a finite winning strategy exactly for

$$\boxed{\theta \in \left\{ \frac{180^\circ}{n} : n = 2, 3, 4, \ldots \right\}.}$$

$\square$

Problem 5: Double-Inequality Functional Equation

Area	Algebra	**Elapsed time**	2m 36s
Transcript	Open shared ChatGPT transcript		
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Problem. *Determine all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that*

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{x f(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Solution

Theorem 1. *The functions satisfying the inequality are exactly*

$$\boxed{f(x) = x + c \qquad (x > 0),}$$

where $c \geq 0$ is an arbitrary constant.

Proof. For $n \geq 0$, let $f^{\circ n}$ denote the n-fold iterate of f, with $f^{\circ 0}$ equal to the identity map. Define

$$d(x) := f(x) - x \qquad (x > 0).$$

1. A basic identity and nonnegativity of d.

Set $x = f(y)$ in the given double inequality. Since $f(y) > 0$, both outer terms are equal to $f(y)$:

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y), \qquad \sqrt{f(y) f(y)} = f(y).$$

Consequently the middle term is also equal to $f(y)$, and hence

$$\frac{f(f(y)) + y}{2} = f(y).$$

Thus

$$f(f(y)) = 2f(y) - y. \tag{1}$$

In terms of d, this gives

$$d(f(y)) = f(f(y)) - f(y) = f(y) - y = d(y). \tag{2}$$

We claim that for every $y > 0$ and every integer $n \geq 0$,

$$f^{\circ n}(y) = y + n d(y). \tag{3}$$

Indeed, the assertion is clear for $n = 0$. If it holds for some n, then (2) implies

$$d(f^{\circ n}(y)) = d(y),$$

and therefore

$$f^{\circ(n+1)}(y) = f^{\circ n}(y) + d(f^{\circ n}(y)) = y + (n+1)d(y).$$

This proves (3) by induction.

Every iterate $f^{\circ n}(y)$ lies in $\mathbb{R}_{>0}$. If $d(y) < 0$, then the right-hand side of (3) is nonpositive for all sufficiently large n, a contradiction. Hence

$$d(y) \geq 0 \qquad \text{for every } y > 0. \tag{4}$$

2. Any two positive values of d are equal.

The first half of the given inequality is equivalent to

$$f(x) + y \leq \sqrt{2(x^2 + f(y)^2)}.$$

Since

$$f(x) + y = x + f(y) + d(x) - d(y),$$

we obtain

$$d(x) - d(y) \leq \Phi(x, f(y)), \tag{5}$$

where, for $r, s > 0$, we set

$$\Phi(r, s) := \sqrt{2(r^2 + s^2)} - r - s.$$

Rationalizing gives the useful identity

$$\Phi(r, s) = \frac{(r-s)^2}{\sqrt{2(r^2+s^2)} + r + s}. \tag{6}$$

Suppose that $d(u) = a > 0$ and $d(v) = b > 0$. By (2) and (3), the points

$$u_n := f^{\circ n}(u) = u + na, \qquad v_m := f^{\circ m}(v) = v + mb$$

satisfy

$$d(u_n) = a, \qquad d(v_m) = b$$

for all $m, n \geq 0$.

For every sufficiently large n, define

$$m_n := \left\lfloor \frac{u_n - v - b}{b} \right\rfloor.$$

Then $m_n \geq 0$ and

$$0 \leq u_n - \big(v + (m_n + 1)b\big) < b. \tag{7}$$

Because $f(v_{m_n}) = v + (m_n + 1)b$, applying (5) with $x = u_n$ and $y = v_{m_n}$ yields

$$a - b \leq \Phi\big(u_n, f(v_{m_n})\big).$$

Using (6) and (7), we find

$$0 \leq \Phi\big(u_n, f(v_{m_n})\big) < \frac{b^2}{u_n + f(v_{m_n})}.$$

Moreover, (7) implies $f(v_{m_n}) > u_n - b$, so the denominator tends to infinity as $n \to \infty$. Therefore

$$\Phi\big(u_n, f(v_{m_n})\big) \longrightarrow 0,$$

and consequently $a - b \leq 0$. Interchanging u and v gives $b - a \leq 0$, so $a = b$.

It follows that either d is identically zero, or there exists a fixed constant $c > 0$ such that

$$d(x) \in \{0, c\} \qquad \text{for every } x > 0. \tag{8}$$

3. The values 0 and c cannot both occur.

Assume now that $c > 0$ occurs, and define

$$Z := \{x > 0 : d(x) = 0\}, \qquad P := \{x > 0 : d(x) = c\}.$$

By (8), the sets Z and P form a partition of $\mathbb{R}_{>0}$. We show that both are open in $\mathbb{R}_{>0}$.

First let $z \in Z$. Then $f(z) = z$. Applying the first half of the given inequality with $y = z$ gives

$$f(x) + z \leq \sqrt{2(x^2 + z^2)},$$

and hence

$$d(x) \leq \sqrt{2(x^2 + z^2)} - x - z. \tag{9}$$

The right-hand side of (9) is a continuous function of x and is equal to 0 at $x = z$. Thus it is strictly smaller than c whenever x is sufficiently close to z. Since $d(x) \in \{0, c\}$, equation (9) then forces $d(x) = 0$. Hence Z is open.

Next let $p \in P$, so $f(p) = p + c$. If $x \in Z$, then $f(x) = x$, and the second half of the given inequality, with $y = p$, would imply

$$\frac{x + p}{2} \geq \sqrt{x(p + c)}. \tag{10}$$

At $x = p$, however, (10) is false, because

$$p < \sqrt{p(p + c)}.$$

By continuity, (10) remains false for every x in some neighborhood of p. No point of that neighborhood can therefore belong to Z. By (8), every such point belongs to P, and hence P is open.

The interval $(0, \infty)$ is connected, so it cannot be written as a union of two disjoint nonempty open subsets. Since P is nonempty, we must have $Z = \varnothing$. Therefore $d(x) = c$ for every $x > 0$. Combining this conclusion with the case $d \equiv 0$, we obtain

$$f(x) = x + c$$

for some constant $c \geq 0$.

4. Verification.

Conversely, let $f(x) = x + c$ for some $c \geq 0$. Then

$$f(y) = y + c \quad \text{and} \quad \frac{f(x) + y}{2} = \frac{x + c + y}{2} = \frac{x + f(y)}{2}.$$

Thus the desired inequalities become

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{x + f(y)}{2} \geq \sqrt{x f(y)},$$

which are exactly the root-mean-square–arithmetic-mean inequality and the arithmetic-mean–geometric-mean inequality, applied to the positive numbers x and $f(y)$. Hence every function $f(x) = x + c$ with $c \geq 0$ is a solution. $\square$

Problem 6: Eventual Translation of a Greedy Sequence

Area	Number theory	**Elapsed time**	15m 10s
Transcript	Open shared ChatGPT transcript		
Navigation	Back to problem index · Gemini review (7/7)		

Problem. *Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that, for every positive integer n, the number a_{n+1} is the smallest integer greater than a_n such that*

$$\gcd(a_{n+1}, a_i) > 1 \qquad (1 \le i \le n).$$

Prove that there exist positive integers T and L such that

$$a_{n+T} = a_n + L$$

for every positive integer n.

Solution

Proof. Define

$$\mathcal{S} := \{x \in \mathbb{Z}_{>0} : \gcd(x, a_i) > 1 \text{ for every } i \ge 1\}.$$

We shall prove that $\mathcal{S}$ is periodic and that the sequence $a_1, a_2, \dots$ lists, in increasing order, precisely the elements of $\mathcal{S}$ that are at least a_1.

First observe that any two terms of the sequence have greatest common divisor larger than 1: when the later term was chosen, it was required to have this property with every earlier term. Hence $a_n \in \mathcal{S}$ for every n.

Because the sequence is strictly increasing, it is unbounded. We claim that

$$\{a_1, a_2, \dots\} = \mathcal{S} \cap [a_1, \infty). \tag{1}$$

The inclusion from left to right was just proved. Conversely, suppose that $x \in \mathcal{S}$, $x \ge a_1$, and x is not a term of the sequence. Since a_1 is a term, we have $x > a_1$; by unboundedness there is an n for which

$$a_n < x < a_{n+1}.$$

The definition of $\mathcal{S}$ gives $\gcd(x, a_i) > 1$ for $1 \le i \le n$, so x was an admissible candidate for a_{n+1}. The minimality of a_{n+1} would then imply $a_{n+1} \le x$, a contradiction. Thus (1) holds.

The set $\mathcal{S}$ is upward closed under divisibility: if $x \in \mathcal{S}$ and $x \mid y$, then $y \in \mathcal{S}$. In particular, if

$$\operatorname{rad}(x) := \prod_{p \mid x} p$$

denotes the squarefree kernel of x, then

$$x \in \mathcal{S} \iff \operatorname{rad}(x) \in \mathcal{S}. \tag{2}$$

Indeed, $\operatorname{rad}(x) \mid x$, and x and $\operatorname{rad}(x)$ have exactly the same prime divisors.

We next record two consequences of the definition of $\mathcal{S}$ and of the greedy rule.

Lemma 1. *Any two elements of $\mathcal{S}$ have a nontrivial common divisor. In other words,*

$$x, y \in \mathcal{S} \implies \gcd(x, y) > 1.$$

Proof. Suppose, to the contrary, that $x, y \in \mathcal{S}$ and $\gcd(x, y) = 1$. Choose a positive integer k such that $k \equiv 1 \pmod{y}$ and $kx \geq a_1$. Because $\mathcal{S}$ is closed under taking multiples, $kx \in \mathcal{S}$. By (1), kx is a term of the sequence. Since $y \in \mathcal{S}$, it must therefore satisfy $\gcd(y, kx) > 1$. But $\gcd(y, k) = \gcd(y, x) = 1$, so $\gcd(y, kx) = 1$, a contradiction. ☐

Lemma 2. *If $x > a_1$ and $x \notin \mathcal{S}$, then there exists a term $a_i < x$ such that*

$$\gcd(x, a_i) = 1.$$

Proof. Since $x \notin \mathcal{S}$, the number x is not a term of the sequence. Thus, by unboundedness, there is an n such that

$$a_n < x < a_{n+1}.$$

If $\gcd(x, a_i) > 1$ held for every $1 \leq i \leq n$, then x would be an admissible candidate for a_{n+1}, contradicting the minimality of a_{n+1}. Hence $\gcd(x, a_i) = 1$ for at least one $i \leq n$, and then $a_i \leq a_n < x$. ☐

Let $\mathcal{M}$ be the set of squarefree elements of $\mathcal{S}$ that are minimal under divisibility. Explicitly,

$$m \in \mathcal{M} \quad \Longleftrightarrow \quad \begin{cases} m \in \mathcal{S}, \quad m \text{ is squarefree,} \\ d \notin \mathcal{S} \text{ for every proper divisor } d \mid m. \end{cases}$$

The set $\mathcal{M}$ is nonempty: by (2), $\operatorname{rad}(a_1) \in \mathcal{S}$, and among the finitely many divisors of $\operatorname{rad}(a_1)$ that lie in $\mathcal{S}$ one may choose a minimal one.

Moreover, every $x \in \mathcal{S}$ is divisible by some member of $\mathcal{M}$. Indeed, by (2) the number $\operatorname{rad}(x)$ lies in $\mathcal{S}$; among its divisors belonging to $\mathcal{S}$, choose one minimal under divisibility. It is squarefree and hence is an element of $\mathcal{M}$. Consequently,

$$\mathcal{S} = \bigcup_{m \in \mathcal{M}} m\mathbb{Z}_{>0}. \tag{3}$$

We now prove that only finitely many primes occur in the members of $\mathcal{M}$. Set $A := a_1$.

Lemma 3 (Descent lemma)**.** *Let $m \in \mathcal{M}$, and let q be a prime divisor of m. Then there is an $m' \in \mathcal{M}$ such that*

$$q \mid m' \qquad \textit{and} \qquad \frac{m'}{q} \leq A.$$

Proof. Start with $m_0 := m$. Suppose that $m_j \in \mathcal{M}$, that $q \mid m_j$, and that $m_j/q > A$. Since m_j is minimal in $\mathcal{S}$, its proper divisor

$$x := \frac{m_j}{q}$$

does not belong to $\mathcal{S}$. The preceding lemma therefore gives a term $a_i < x$ such that $\gcd(a_i, x) = 1$.

By (2), $\operatorname{rad}(a_i) \in \mathcal{S}$. Choose $m_{j+1} \in \mathcal{M}$ dividing $\operatorname{rad}(a_i)$. Then

$$m_{j+1} \leq \operatorname{rad}(a_i) \leq a_i < x = \frac{m_j}{q} < m_j, \tag{4}$$

and, because $m_{j+1} \mid \operatorname{rad}(a_i)$, we also have $\gcd(m_{j+1}, x) = 1$.

Both m_j and m_{j+1} lie in $\mathcal{S}$, so the first lemma yields $\gcd(m_j, m_{j+1}) > 1$. Now m_j is squarefree and $m_j = qx$, while m_{j+1} is coprime to x. It follows that the only possible common prime divisor is q. Hence $q \mid m_{j+1}$.

Thus, whenever $m_j/q > A$, we can replace m_j by a strictly smaller element $m_{j+1} \in \mathcal{M}$ that is still divisible by q. This descent cannot continue indefinitely. At its terminal stage we obtain an $m' \in \mathcal{M}$ with $q \mid m'$ and $m'/q \leq A$, as required. ☐

Let P be the set of primes that divide at least one member of $\mathcal{M}$. Suppose for contradiction that P is infinite. For each $q \in P$, the descent lemma supplies an element

$$m_q = qc_q \in \mathcal{M} \qquad \text{with} \qquad 1 \le c_q \le A.$$

Because c_q is a proper divisor of the minimal element $m_q \in \mathcal{M}$, we have $c_q \notin \mathcal{S}$.

There are only finitely many possible values of c_q. Hence there is a fixed integer $c \in \{1, 2, \dots, A\}$ such that

$$qc \in \mathcal{M} \tag{5}$$

for infinitely many distinct primes q. For this c we have $c \notin \mathcal{S}$, so by the definition of $\mathcal{S}$ there is some term a_j with

$$\gcd(c, a_j) = 1.$$

For each prime q satisfying (5), both a_j and qc belong to $\mathcal{S}$. The first lemma therefore gives

$$\gcd(a_j, qc) > 1.$$

Since a_j is coprime to c, this forces $q \mid a_j$. Thus the fixed positive integer a_j is divisible by infinitely many distinct primes, an impossibility. We conclude that P is finite.

Every member of $\mathcal{M}$ is squarefree and all of its prime divisors lie in the finite set P. Hence $\mathcal{M}$ itself is finite. Define

$$L := \prod_{p \in P} p.$$

Every $m \in \mathcal{M}$ divides L. Using (3), we therefore obtain, for every $x \in \mathbb{Z}_{>0}$,

$$\begin{aligned} x \in \mathcal{S} &\iff m \mid x \text{ for some } m \in \mathcal{M} \\ &\iff m \mid x + L \text{ for some } m \in \mathcal{M} \\ &\iff x + L \in \mathcal{S}. \end{aligned}$$

Thus membership in $\mathcal{S}$ is periodic with period L.

Let

$$T := \big|\mathcal{S} \cap \{1, 2, \dots, L\}\big|.$$

This number is positive: $\mathcal{M}$ is nonempty, every $m \in \mathcal{M}$ divides L, and $\mathcal{S}$ is closed under taking multiples, so $L \in \mathcal{S}$. Periodicity implies that every interval of L consecutive integers contains exactly T elements of $\mathcal{S}$. In particular, the interval

$$(a_n, a_n + L]$$

contains exactly T elements of $\mathcal{S}$, and its right endpoint belongs to $\mathcal{S}$ because $a_n \in \mathcal{S}$.

By (1), the sequence lists the elements of $\mathcal{S} \cap [a_1, \infty)$ in strictly increasing order. Hence the T-th term after a_n is exactly the last element of $\mathcal{S}$ in $(a_n, a_n + L]$, namely $a_n + L$. Therefore

$$\boxed{a_{n+T} = a_n + L}$$

for every positive integer n. $\square$

A Reviews

A.1 Problem 1: GCD–LCM Blackboard Process— Review by Gemini

Return to Problem 1: GCD–LCM Blackboard Process.

Overall Impression

This is an exceptionally well-crafted, rigorous, and elegant solution to a mathematical olympiad problem. The logic is sound, and despite applying extreme skepticism, I could find no mathematical gaps, missing edge cases, or flaws in the reasoning.

Critique of Part (a): Termination

The author beautifully employs the method of monovariants (decreasing functions) to prove termination. By defining T as the total sum of prime exponents and R as the number of non-trivial entries, the author constructs the non-negative integer function $\Phi = T + R$.

The algebraic verification that $T_{\text{old}} - T_{\text{new}} = \Omega(d)$ is rigorously established using the properties of p-adic valuations, specifically that $v_p(d) = \min(a, b)$ and $v_p(q) = |a - b|$. The case-work cleanly separates the dynamics:

- If $d > 1$, the total prime factor count T strictly decreases, and R cannot increase.
- If $d = 1$, T remains unchanged, but the two selected entries are replaced by exactly one non-trivial entry, meaning R strictly decreases.

Because Φ must strictly decrease after every move, the process is guaranteed to terminate. The deduction that exactly one element $M > 1$ remains is also perfectly argued; R can never drop to zero because any move generates at least one number greater than 1, and the game only stops when no two elements > 1 are left.

Critique of Part (b): Invariance

The construction of the invariant $I_p = \gcd(v_p(x_1), \ldots, v_p(x_{2026}))$ is the most elegant aspect of this proof. The author accurately observes that replacing the valuations a and b with $\min(a, b)$ and $|a - b|$ mirrors a single step of the Euclidean algorithm. This seamlessly proves that I_p is strictly conserved.

Because the terminal board state consists of a single integer M alongside ones, the final state of the invariant reduces to $v_p(M)$. Equating this final state to the initial state proves that every exponent in the prime factorization of M is predetermined solely by the initial board.

Final Verdict

Score: 7/7

This is a textbook-quality, perfect solution. The proof correctly isolates the necessary monovariant and invariant with precise, standard notation. It leaves no room for ambiguity, requires no logical leaps, and handles conventions like $\gcd(0, t) = t$ correctly.

A.2 Problem 2: Equal Midpoint Distances from a Circumcentre— Review by Gemini

Return to Problem 2: Equal Midpoint Distances from a Circumcentre.

Overall Verdict: 7/7

The provided solution is rigorous, mathematically sound, and merits a perfect score of 7 out of 7 under standard International Mathematical Olympiad (IMO) grading criteria.

Detailed Review

Applying a high level of mathematical skepticism to the proof reveals a meticulously structured argument that correctly anticipates and addresses standard pitfalls in olympiad-level geometry.

1. Configuration and Topology: The author commendably establishes the exact topological ordering of the rays from vertex A. By utilizing positive barycentric coordinates and exploiting the monotonic properties of ray distributions, the proof rigorously guarantees the geometric configuration AB, AK, AL, AC. This successfully seals the notorious configuration gaps that frequently compromise synthetic geometry submissions.

2. Algebraic Translation: The proof leverages the two midpoint conditions flawlessly. By translating synthetic angle relations into the sine rule, it effectively bridges the geometric layout into a system of trigonometric constraints. There are no unaddressed division-by-zero errors; all denominator terms (such as X, Y, Z, P, Q) safely correspond to sines of strictly positive, non-degenerate angles strictly bounded by π.

3. Directed Angles and Power of a Point: The transition to evaluating the power of points M and N with respect to the circumcircle ω of $\triangle AKL$ is handled with precision. The usage of directed angles modulo π safely sidesteps sign and orientation errors when applying the secant form of the power theorem. The respective derivations of $\mathrm{Pow}_\omega(M)$ and $\mathrm{Pow}_\omega(N)$ are completely accurate.

4. Trigonometric Reduction: The concluding section relies on complex trigonometric manipulation to prove the equivalence condition $D = 0$. Rather than brute-forcing a blind expansion, the author isolates the S-dependent terms into an auxiliary function $F(S)$. Through correct application of product-to-sum formulas, they establish that $F(S)$ is independent of S. Evaluating this constant function at a strategically chosen $S_0 = x + y + z$ successfully collapses the entire expression to zero.

Commentary on Elegance

While purists might argue that a purely synthetic geometry approach—perhaps utilizing spiral similarities, inversion, or projective transformations—feels more "elegant," this trigonometric bash is a structural masterclass. The author does not simply throw equations at the wall; they systematically reduce a notoriously difficult geometric configuration into a highly controlled, deterministic algebraic identity. It demonstrates immense algebraic fortitude, leaving virtually no room for logical loopholes or unproven geometric assumptions.

A.3 Problem 2: Equal Midpoint Distances from a Circumcentre— Additional Jury-Style Review

Return to Problem 2: Equal Midpoint Distances from a Circumcentre.

Review provenance. This additional review was prepared with assistance from ChatGPT 5.6 Pro and adopted by the authors as the basis for their revised assessment of Problem 2.

Overall Assessment

The main strategy appears sound, and the later barycentric and trigonometric manipulations may be correct. However, an IMO jury would likely regard the power calculation in Section 3 as insufficiently rigorous.

Points Requiring Justification

1. Directed-angle sign. The proof uses directed angles and the sine rule to obtain

$$\frac{\overline{MT}}{MA} = \frac{\sin(z-\lambda)}{\sin\lambda}.$$

This requires a precise convention linking representatives modulo π to directed lengths. An angle known only modulo π does not have a well-defined sine, because $\sin(\alpha+\pi) = -\sin\alpha$. Moreover, under a natural orientation, the stated congruence for $\angle AMT$ appears to have the opposite sign. The formula may be true, but it is not established as written. A short signed-coordinate computation or an explicit directed-sine lemma would repair the gap.

2. Nondegeneracy of the second intersections. Writing "let $T \neq K$ be the second intersection of MK with ω" assumes that MK is not tangent to ω at K. Applying the sine rule in triangle AMT also assumes $T \neq M$. Thus tangency and the possibility $M \in \omega$ must be excluded; the analogous issues occur for N, L, U. These facts should be proved before the power calculation, or intersections should be counted with multiplicity.

3. Positivity and formal substitution. The proof relies on strict barycentric inequalities and divides by several sine factors, so strict interiority and nonvanishing denominators should be recorded explicitly. When S is later varied and a special value S_0 is substituted, the argument should also state that this is a formal trigonometric identity, not a geometric deformation, and verify the relevant denominators.

4. Notation. The symbol T denotes both a point and later a scalar expression; one should be renamed.

Score Discussion

IMO grading does not impose an automatic one-point deduction for every omission. A genuinely routine and unambiguous detail may be forgiven, sometimes even with a score of 7. Here, however, the issue lies in a central power computation, includes a possible sign inconsistency, and uses unproved nondegeneracy. A perfect score would therefore be difficult to justify without the missing argument.

A fair score is

$$\boxed{6/7}.$$

The main method remains intact and the gap appears locally repairable, so a large deduction would be excessive. A score of 5/7 would also be defensible under a stricter marking scheme, but 6/7 is the most balanced assessment.

Final Verdict

The solution is essentially correct in conception but not fully rigorous as submitted. It deserves substantial credit, but not 7/7, until the directed-sign calculation and the degenerate-intersection cases are handled explicitly.

We note that the shortfall was due to one-shot nature of this assesment only. It could easily be fixed with better initial prompt or additional review passes.

A.4 Problem 3: Mark-and-Claim Stick Game— Review by Gemini

Return to Problem 3: Mark-and-Claim Stick Game.

Correctness and Gap Analysis

I reviewed the provided proof with intense skepticism, looking specifically for unwarranted assumptions in the game-theoretic formulation and the greedy algorithm. The argument is entirely rigorous and gap-free.

- **Claiming Phase (Lemma 1):** The lemma relies on the greedy strategy (always taking the largest available piece). For a general set of independent positive numbers, this strategy is demonstrably optimal, and the alternating sum exactly determines the first player's advantage. The inductive bounding argument is airtight.
- **Upper Bound:** The application of the Pigeonhole Principle to the 2^{n+1} subset sums is flawless. By bounding the gap between two subset sums by $u = 1/(2^{n+1} - 1)$, Xiang Yu's strategy guarantees the residual unmatched length is at most u. The cut-counting in the matching lemma is completely rigorous. The strict constraint of utilizing at most n cuts is respected.
- **Lower Bound:** The choice of geometric initial lengths $u, 2u, \ldots, 2^n u$ is strategically sound. The potential gap in such proofs usually lies in proving the opponent cannot shatter the geometric dominance. The graph-theoretic approach to bound the alternating sum avoids this pitfall entirely. By constructing a multigraph with at most n edges on $n + 1$ vertices, the existence of a tree component is structurally guaranteed. The bipartite coloring of this tree perfectly tracks the alternating signs, ensuring the largest power of 2 strictly dominates the sum of the smaller powers.

Commentary on Elegance

This is an exceptionally elegant solution to a highly complex combinatorial game. The proof expertly weaves together continuous-interval matching, the Pigeonhole Principle, and algebraic graph theory. The transition from a continuous cutting game into a discrete graph problem in the lower bound is a masterstroke, neatly bypassing what would otherwise be a messy casework-heavy algebraic slog.

Final Verdict

Based on standard International Mathematical Olympiad (IMO) grading criteria, this solution contains no logical gaps, properly addresses both the upper and lower bounds of the game value, and correctly formalizes the optimal strategies for both players.

Score: 7/7

A.5 Problem 4: Triangle-Cutting Angle Game— Review by Gemini

Return to Problem 4: Triangle-Cutting Angle Game.

Skeptical Analysis and Verification

We approach this solution with intense skepticism, probing for any gaps in the logical flow or unstated assumptions, particularly concerning geometric configurations and algebraic invariant arguments.

Sufficiency ($\theta = 180°/n$): The author's strategy hinges on a preliminary claim that creating an angle of $m\theta$ leads to a win in at most $m-1$ further cuts. The induction here is flawless because a ray dividing an interior angle of $m\theta$ into θ and $(m-1)\theta$ is always geometrically realizable and strictly internal to the triangle.

The core of the sufficiency proof relies on a custom Lemma to find an integer k such that $b < k < a+b$. We verified the bounding logic: 1. If $a > 1$, setting $k = \lceil b \rceil$ guarantees $b < k < b+1 < b+a$, which holds strictly because b is a noninteger. 2. If $a, b, c < 1$ and $a+b+c = n \geq 2$, then n must be exactly 2. The largest two elements a, b must sum to strictly greater than 1, making $k = 1$ a valid choice.

This ensures Mulan's proposed cut angle of $(k-b)\theta$ is strictly positive and less than $a\theta$, proving it is a legal cut. Calculating the newly formed angles, $(n-k)\theta$ and $k\theta$, confirms that both resulting triangles contain a positive integer multiple of θ. There are no topological or algebraic gaps here; Shan-Yu is forced to hand Mulan a winning state.

Necessity ($\Omega/\theta \notin \mathbb{Z}$): The construction of the additive subgroup $H = \{m\theta \mid m \in \mathbb{Z}\}$ is a standard but powerful invariant tool. Shan-Yu choosing an equilateral triangle is legally permitted by the problem statement, which gives him full initial choice. The verification of the invariant under an arbitrary cut $x \in (0, a)$ rigorously checks all four possible failure states. The algebraic contradictions are meticulously and correctly derived: $x + (a-x) = a \notin H$, $(b+x) - x = b \notin H$, $(\Omega - b - x) - (a-x) = c \notin H$, and $(\Omega - b - x) + (b+x) = \Omega \notin H$. Since none of these pairings can exist, at least one triangle is entirely free of angles in H. The logic is perfectly airtight.

Commentary and Elegance

This is a remarkably elegant, complete, and well-written solution. The author avoids messy geometric edge cases by elegantly lifting the problem into algebraic constraints (the $a+b+c=n$ vector space for sufficiency, and the additive subgroup H for necessity). The transition from geometric cuts to linear combinations of angles is seamless. The lemma in the sufficiency proof is particularly beautiful, effortlessly turning a continuous geometric obstacle into a discrete arithmetic bound.

Verdict

IMO Score: 7/7
The proof is mathematically rigorous, impeccably structured, and completely resolves the problem without any logical leaps or missing edge cases.

A.6 Problem 5: Double-Inequality Functional Equation— Review by Gemini

Return to Problem 5: Double-Inequality Functional Equation.

Logical Rigor and Skeptical Analysis

Approaching this proof with strict skepticism reveals a highly rigorous, watertight argument with no logical gaps. The author tackles a potentially messy functional inequality using a remarkably clear, four-step structure.

- **Step 1 (Basic Identity):** The initial substitution $x = f(y)$ perfectly exploits the structure of the given double inequality. Squeezing the middle term to force $f(f(y)) = 2f(y) - y$ is algebraically sound. The subsequent inductive claim, $f^{\circ n}(y) = y + nd(y)$, correctly deduces the non-negativity of $d(y)$ by relying on the strictly positive codomain of the function.
- **Step 2 (Constraining d):** This is the most analytically delicate part of the proof. The author introduces $\Phi(r, s)$ to cleanly isolate $d(x) - d(y)$. The construction of the arithmetic sequences u_n and v_{m_n} successfully avoids the common pitfall of assuming the function is continuous or monotonic. The bounding of the difference $0 \le u_n - (v + (m_n + 1)b) < b$ is strictly verified. This allows the limit $\Phi \to 0$ as $n \to \infty$ to rigorously prove that any two positive values of d must be equal.
- **Step 3 (Topological Conclusion):** Reducing the problem to $d(x) \in \{0, c\}$ is a strong milestone. Defining Z and P as partitions requires finesse to prove they cannot coexist. The use of topology here is flawless. The author successfully leverages the strict inequality $p < \sqrt{p(p+c)}$ and the continuity of the bounding equations to show both Z and P are open subsets. Appealing to the connectedness of $(0, \infty)$ to eliminate the partition completely forces the function to be a constant shift.
- **Step 4 (Verification):** The verification step is complete. It correctly reduces the substituted function to the classic RMS-AM-GM inequalities.

Quality and Elegance

This is a sophisticated, competition-level solution. It masterfully blends algebraic manipulation, asymptotic bounding, and point-set topology. The deliberate avoidance of analytical assumptions in the early steps, followed by a topological finish to classify the sets, is highly elegant and creative.

Final Verdict

Score: 7 / 7

The solution is entirely complete, rigorously justified at every step, and exceptionally well-written. It meets the highest standards of international olympiad grading criteria.

A.7 Problem 6: Eventual Translation of a Greedy Sequence— Review by Gemini

Return to Problem 6: Eventual Translation of a Greedy Sequence.

Final Verdict

Score: 7/7

The provided proof is exceptionally rigorous, logically airtight, and demonstrates a high degree of mathematical maturity. After a thoroughly skeptical review, no logical gaps, circular reasoning, or unverified assumptions were found. It warrants a perfect score under standard International Mathematical Olympiad (IMO) grading criteria.

Detailed Analysis of the Argument

The author constructs the proof around the properties of the set of eventual sequence multiples, $\mathcal{S}$.

- **Set $\mathcal{S}$ and Asymptotic Equality:** The assertion that the sequence eventually enumerates $\mathcal{S}$ is flawlessly deduced from the greedy minimality condition. If an element in $\mathcal{S}$ were skipped, it would be a valid candidate and violate the minimality of a_{n+1}.

- **Lemma 1 (Mutual non-coprimality):** The proof uses a standard but elegant modular arithmetic construction, picking a multiple $k \equiv 1 \pmod{y}$ with $kx \geq a_1$, to force a contradiction if any two elements in $\mathcal{S}$ are coprime. This step is perfectly valid.

- **Lemma 2 and the Descent Lemma:** This is the theoretical engine of the proof. By defining $\mathcal{M}$ as the minimal squarefree elements of $\mathcal{S}$, the author sets up an infinite descent on the prime divisors. If a prime q divides $m \in \mathcal{M}$ such that $m/q > a_1$, the proof leverages Lemma 2 to find a smaller element m' still divisible by q. The deduction that q must be the shared prime factor between m_j and m_{j+1} is highly precise and correctly relies on Lemma 1.

- **Finiteness of Prime Divisors:** The pigeonhole principle is applied masterfully here. Since the cofactors c_q are bounded by a_1, infinitely many primes would force a single term a_j to have infinitely many prime divisors, yielding a strict contradiction.

- **Periodicity:** With the set of primes P being finite, $\mathcal{M}$ is finite. The definition of L as the product of these primes cleanly establishes the periodicity of $\mathcal{S}$, finishing the proof.

Commentary and Elegance

The solution is remarkably elegant. Instead of attempting to track the chaotic, step-by-step arithmetic behavior of the greedy sequence, the author steps back to analyze the global algebraic structure of the sequence's "admissible pool." By demonstrating that this pool acts similarly to an ideal governed by a finite basis of squarefree integers, the problem is elegantly reduced from sequential dynamics to static set properties. The notation is clean, the lemmas are perfectly scoped, and the logical transitions are seamless.

B Prompts Used

Common prompt template was used for each problem. The web search capability was disabled from ChatGPT settings as well as instructed in the prompt.

B.1 Solver prompt (ChatGPT 5.6 Pro)

Can you use deep think for at least 60 minutes and use any means at your disposal (except web search), to solve the following hard math problem? Try multiple ideas even if you reach dead ends on some of them. If you find a solution idea, try to formalize into a rigorous solution, and skeptically review it for any gaps or bugs. Do not present partial progress or approaches, only provide a complete rigorous verified solution.

<Actual problem statement>

B.2 Writer prompt (ChatGPT 5.6 Pro)

Can you generate a LaTeX source for the full solution written up properly?

B.3 Reviewer prompt (Gemini)

Can you review this claimed solution on a hard math problem? Be extremely skeptical and try to find any gaps in the argument or flaws in the proof. Give a final verdict on the score between 0 and 7, use typical grading criteria of the International Mathematical Olympiad.

Provide some commentary on the quality or elegance of the solution, or any other observation. Make this review relatively short (500 words or less) and give me a LaTeX source for the review rather than usual rendered response.

B.4 Additional Problem 2 review (ChatGPT 5.6 Pro-assisted)

The additional review of Problem 2 was developed through a short multi-turn exchange. The core requests were to search skeptically for implicit assumptions involving point and angle order, signs, degeneracies, and extreme configurations; to assess the solution under typical IMO grading criteria on a scale from 0 to 7; and to condense the resulting jury-style assessment to at most 500 words.